LF054 ARM7TDMI - TSMC 0.18um 1.8V/3.3V 1P6M (CL018G) Process

(Revision r3p0)

Core Configuration and Performance Summary

GENC-006013 1.0

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Change history

Issue	Date	By
1.0	16 March 2005	Luke Durrant

Contents

1 ABOUT THIS DOCUMENT

1.1 Scope

The purpose of this document is to provide configuration and performance summary data for an ARM7TDMI macrocell implemented on the TSMC 0.18um 1.8V/3.3V 1P6M Process.

1.2 References

This document refers to the following documents.

Ref	Doc No	Title
1	ARM DII 0101	ARM7TDMI Integration Manual
2	ARM DDI 0029F	ARM7TDMI Technical Reference Manual

1.3 Terms and abbreviations

This document uses the following terms and abbreviations.

Term	Meaning
ASIC	Application Specific Integrated Circuit
EDA	Electronic Design Aid
I/O	In / Out
RISC	Reduced Instruction Set Computer
SI	Signal Integrity

2 INTRODUCTION

2.1 Hard Macrocell

The ARM7TDMI core is a 32-bit embedded RISC processor delivered as a hard macrocell optimized to provide the best combination of performance, power and area characteristics. The ARM7TDMI core enables system designers to build embedded devices requiring small size, low power and high performance.

For more information about the ARM7TDMI macrocell please refer to Doc Ref 2.

For more information about how to integrate the ARM7TDMI core into your design please refer to Doc Ref 1.

3 CORE CONFIGURATION

3.1 Core configuration

Source revision	ARM7TDMI r3p0
Module name	ARM7TDMI
Clock gating	No

Number of metal layers used	3 (pattern fill on M4)
Layers used for power rails	1 and 3 M1 (Left & Right), M3 (Left & Right)
I/O connection metal layer used	Vdd/Vss: 1 and 3. Signal: 2 and 3
I/O connection sides used	Core interface: Left, Right and Top. Vdd/Vss: Left and Right.

4 PROCESS

4. Foundry process

Silicon vendor	TSMC
Process name/geometry	CL018G 0.18um 1.8V/3.3V 1P6M Process

Process condition	**Voltage**	**Temperature**
Slow (worst case)	1.62V	+125 °C
Typical	1.80V	+25 °C
Fast1	1.98V	0 °C
Fast2 (best case)	1.98V	-40 °C

5 PERFORMANCE

5.1 Maximum frequency and hold margin

Tools: Note that Clock frequency, skew and other timing information derived from transistor-level analysis using: Pathmill version 5.5_p2

Nanosim version 2002.03-SP2

Maximum operating frequency refers to internal (register-to-register) timing. No SI effects are taken into account for the figures below:

Maximum operating frequency (worst case conditions)	89MHz
Internal hold margin (best case conditions)	159ps

Maximum operating frequency (Typical Conditions)	151MHz
Maximum operating frequency (Fast1 Conditions)	223MHz
Maximum operating frequency (Fast2 Conditions)	234MHz

5.2 Dimensions and area

Area of core (mm^2)	0.53440
Size of core (w x h mm)	0.91485(width) x 0.58415(height)
Aspect ratio (h/w)	1.5661

5.3 Clock latency data

MCLK Port				
	Slow	**Typical**	**Fast1**	**Fast2**
Longest latency	3.400ns	2.019ns	1.433ns	1.358ns

5.4 Power & Voltage (IR) drop analysis

Total Power:	**Slow**
Conditions:	1.62V, 125C, Slow Spice models
Dynamic power @1MHz, Dhrystone	0.173mW
Static power (leakage)	Tbd
Analysis methodology: Transistor-level analysis using Nanosim version 2002.03-SP2	

Total Power:	**Typical**
Conditions:	1.80V, 25C, Typical Spice models
Dynamic power @1MHz, Dhrystone	0.241mW
Static power (leakage)	Tbd
Analysis methodology: Transistor-level analysis using Nanosim version 2002.03-SP2	

Total Power:	**Fast1**
Conditions:	1.98V, 0C, Fast Spice models
Dynamic power @1MHz, Dhrystone	0.356mW
Static power (leakage)	tbd
Analysis methodology: Transistor-level analysis using Nanosim version 2002.03-SP2	

Total Power:	**Fast2**
Conditions:	1.98V, -40C, Fast Spice models
Dynamic power @1MHz, Dhrystone	0.349mW
Static power (leakage)	tbd
Analysis methodology: Transistor-level analysis using Nanosim version 2002.03-SP2	

IR drop:	
Conditions	1.8V, 25°C, 157 MHz, 40% global toggle rate
Average VDD Voltage drop	tbd
Average VSS Voltage rise	tbd
Combined	tbd
Analysis methodology: Transistor-level analysis using Voltagestorm version tbd	

5.5 Test Coverage

Using the full set of ARM7TDMI production vectors (Either in their serial (JTAG) or parallel (AMBA) form), the fault coverage of the ARM7TDMI has been measured at approximately 95%.